

TimeLightHub — Executive Summary

A composable photonic computer that thinks in light

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The problem. Computing is hitting a wall made of heat and energy. A single high-end AI chip draws ~700 watts, most of it spent not on computing but on *moving data* and fighting heat. Memory leaks power just to remember. The industry needs a fundamentally more efficient way to compute — not a smaller transistor.

The idea. TimeLightHub computes with **light** instead of electricity, and remembers with light that **needs no power**. It is built from thirteen photonic elements that compose — like the sections of an orchestra — into a CPU, an AI engine, or a search engine, all from the same hardware, reconfigured by color.

Why it wins — energy. Electronic chips pay four taxes every second: charging capacitors, heating wires, refreshing memory, and moving data. TimeLightHub's compute path pays none of them. Light computes without charging capacitors or heating wires; its memory holds data with no power; and in its AI engine the model's **weights never move** — they live where the light reads them, eliminating the single largest energy cost in artificial intelligence. The whole claim reduces to one measurable number — the energy to read one stored weight — and published photonic physics puts the achievable window at **tens of times less energy per answer**. Our first experiment measures exactly that number.

What's new.

- A **non-volatile memory made of light** — keeps data with the power off, designed as a composable element of a computer (the storage physics is demonstrated science; the latch element is ours).
- An **AI engine where the weights never move** — answers each question for a fraction of the energy.
- **Self-correcting computation** — a three-fold element votes out errors, the way spacecraft computers do.
- A **composable element kit** — one device, many machines, reconfigured by light, with its own language (LUMINA) — every element honestly marked: demonstrated, grounded in

demonstrated physics, or our new design.

Honest status. This is a complete architecture whose every element rests on published science. It is not yet built, and two engineering questions (analog precision at depth; the amplifier pump budget) are stated openly, each with a designed answer and a stage that measures it. The entire design reduces to **one decisive first experiment**: build a single light-switchable memory cell and measure how fast and how cheaply it reads and writes — **a modest, well-bounded first experiment — months, not years; materials-scale, not program-scale; publishable on its own; budget scoped with the partner.** That result turns the design into evidence.

The market. Photonic computing is real and well funded (Lightmatter ~\$850M raised; Akhetonics, Celestial AI, Ayar Labs active). Every funded player builds *interconnects* or a *CPU*; **none pairs a non-volatile optical memory with a composable compute engine and a language.** That is TimeLightHub's space.

The ask. A photonics laboratory or foundry partner to build the first cell and node, and **support for Phase 1** — which carries the program, with labor, from the first cell through the first working node, scoped together with the partner. The intellectual property is protected; the deeper engineering is ready to share under agreement.

The vision. A computer of a few dozen parts instead of billions — that never loses your data, barely warms, corrects its own errors, and answers each question for a fraction of the energy. The science is real, the design is complete; the only question left is *who builds the first one*.

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